

Unit 6, Lesson 5: Drawing Transformations



Benchmark

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformation on a coordinate plane.



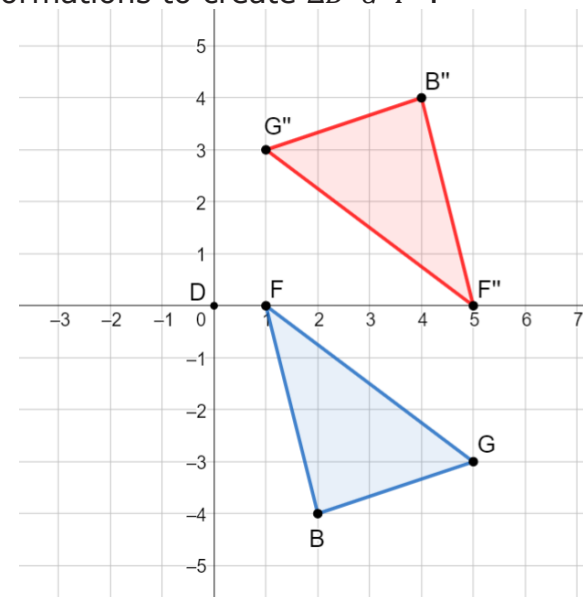
Learning Target

- ☐ I can draw the transformed figure when given a geometric figure and a sequence of transformations as a written description or an algebraic description.



6.5.1 Warm-Up: Error Analysis

- $\triangle BGF$ was transformed by a sequence of transformations to create $\triangle B''G''F''$.



Brian and Emily were determining the sequence of transformations and wrote their answers in the table.

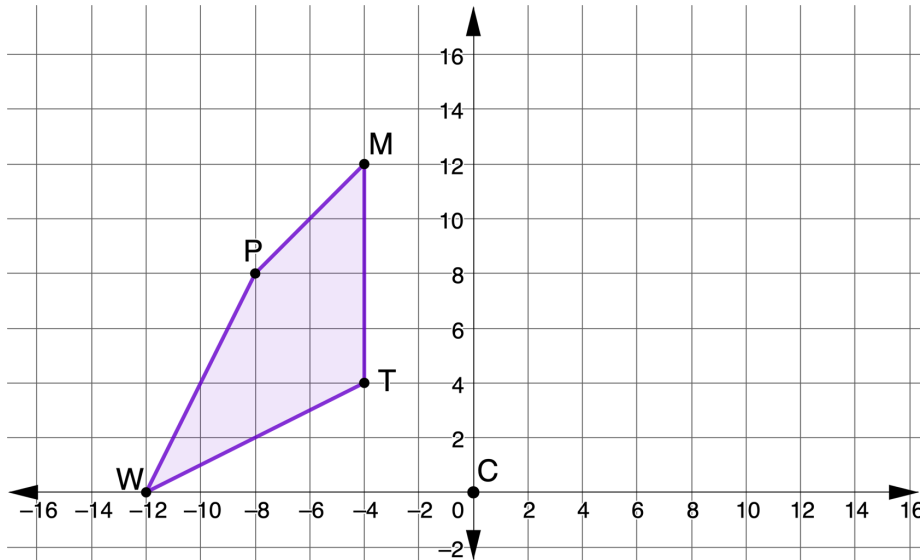
	$\triangle BGF$ Transformed to $\triangle B'G'F'$	$\triangle B'G'F'$ Transformed to $\triangle B''G''F''$
Brian's Answer	Rotated 180° about the origin	Translated right 6 units
Emily's Answer	Reflected over the x -axis	Translated right 4 units

Determine whose sequence of transformations is correct. Explain your reasoning.



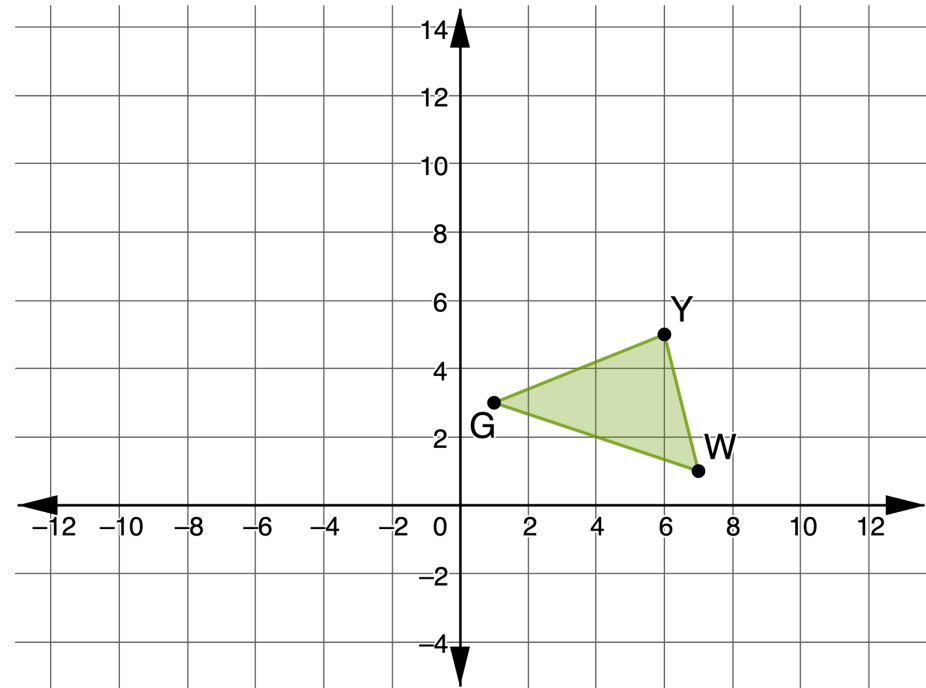
6.5.2 Guided Instruction: Drawing Transformations

1. $MTWP$ and point C are shown on the coordinate plane.



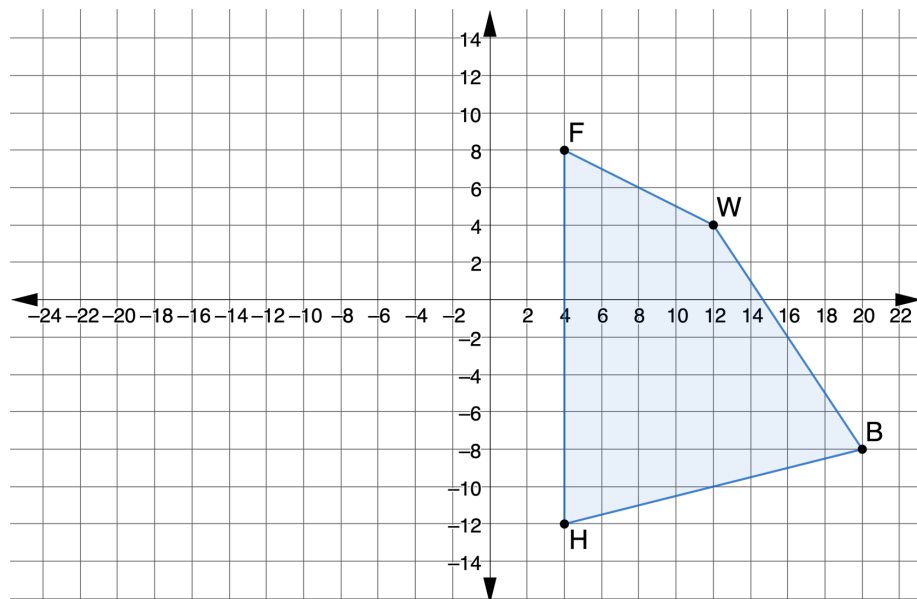
- Draw the quadrilateral that is a reflection of $MTWP$ over the y -axis on the coordinate plane above. Label the quadrilateral $M'T'W'P'$.
- Draw the quadrilateral that is a dilation of $M'T'W'P'$ centered at point C using a scale factor of $\frac{1}{4}$ on the coordinate plane above. Label the quadrilateral $M''T''W''P''$.
- Compare your drawings with your partner. Make any corrections if needed.

2. $\triangle GYW$ is shown on the coordinate plane.



- Draw $\triangle G'Y'W'$ as a result of $(x, y) \rightarrow (x - 4, y - 1)$ on the coordinate plane.
- Draw $\triangle G''Y''W''$ as a result of $(x, y) \rightarrow (3x, 3y)$ on the coordinate plane.

3. $FWBH$ is shown on the coordinate plane.

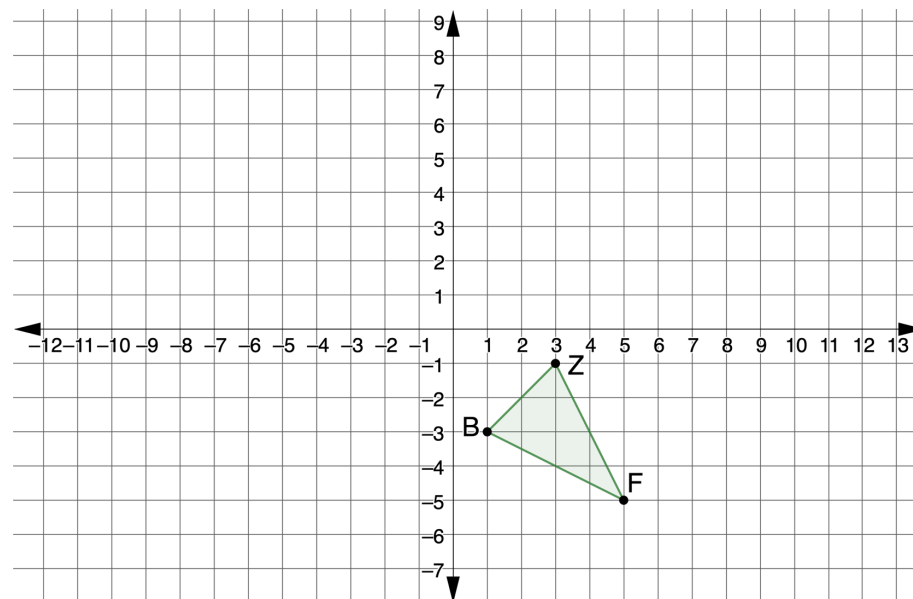


On the same coordinate plane, draw $F''W''B''H''$, which is a result of dilating $FWBH$ using a scale factor of $\frac{3}{4}$, with the center at point F , then rotating the quadrilateral 90° clockwise about point F .



6.5.3 Guided Practice: Sequence of Transformations

1. $\triangle FZB$ is shown on the coordinate plane.

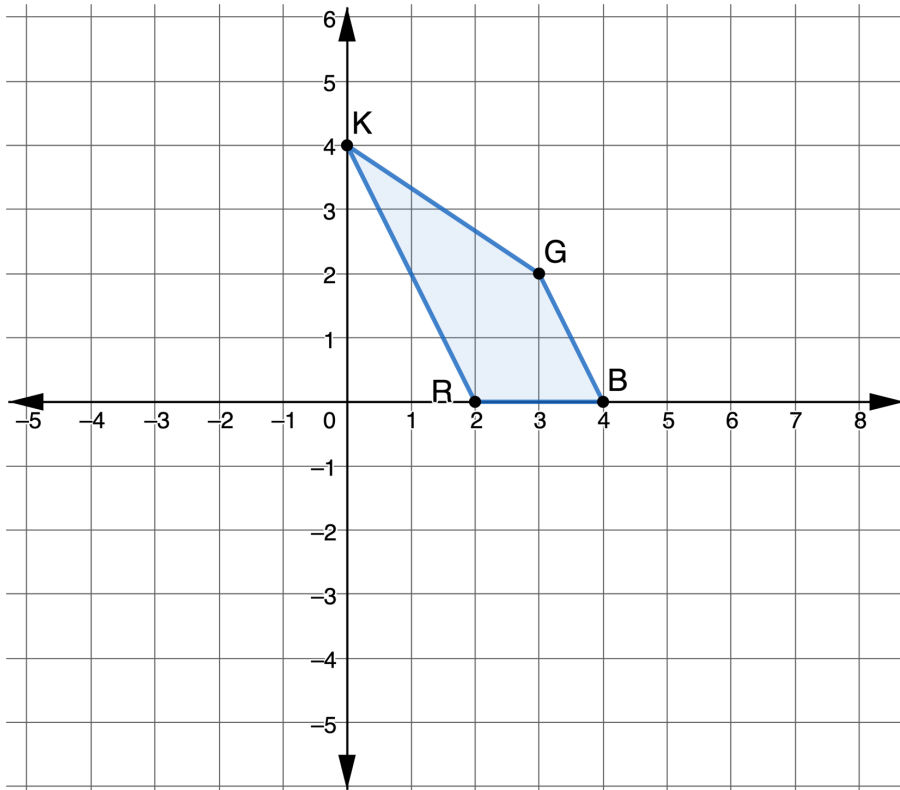


On the same coordinate plane, draw $\triangle F'''Z'''B'''$, which is the result of a reflection over the x -axis, then translated left 3 units and down 3 units, then dilated with a scale factor of $\frac{5}{2}$ using point Z'' as the center of dilation.



6.5.4 Your Turn!

1. $KGBR$ is shown on the coordinate plane.



On the same coordinate plane, draw $K''G''B''R''$, which is the result of $(x, y) \rightarrow (y, -x)$, then $(x, y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$.



6.5.5 Wrap-Up: Cool Down

1. Why do you think it is important to use the prime notation when working with transformations?